



BIO264

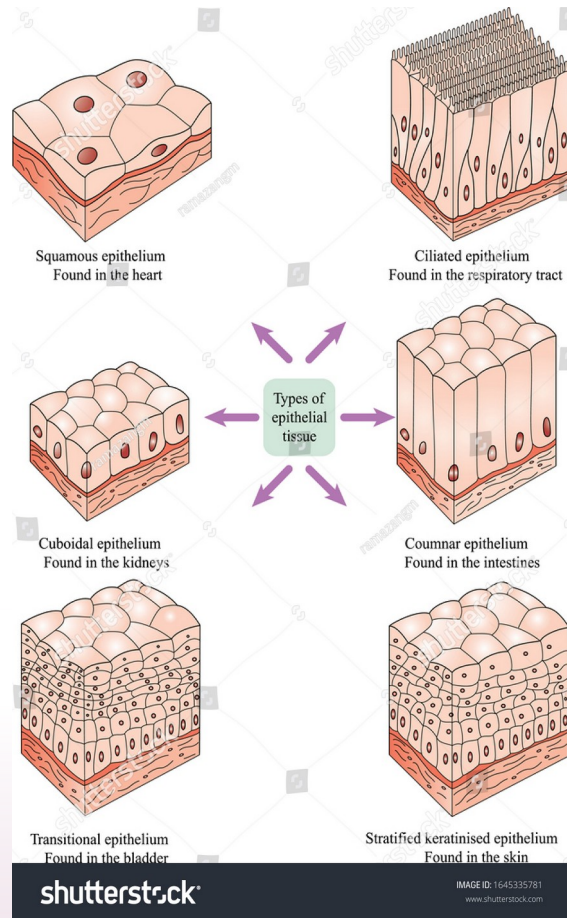
PHYSIOLOGY

Module 3

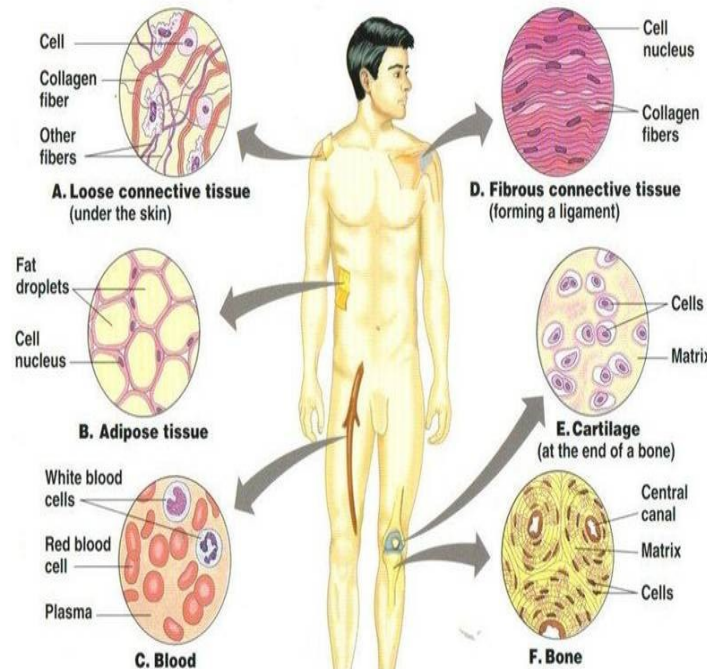
- Tissues and membranes
- Skin/integumentary

HISTOLOGY - 4 MAIN TYPES TISSUES:

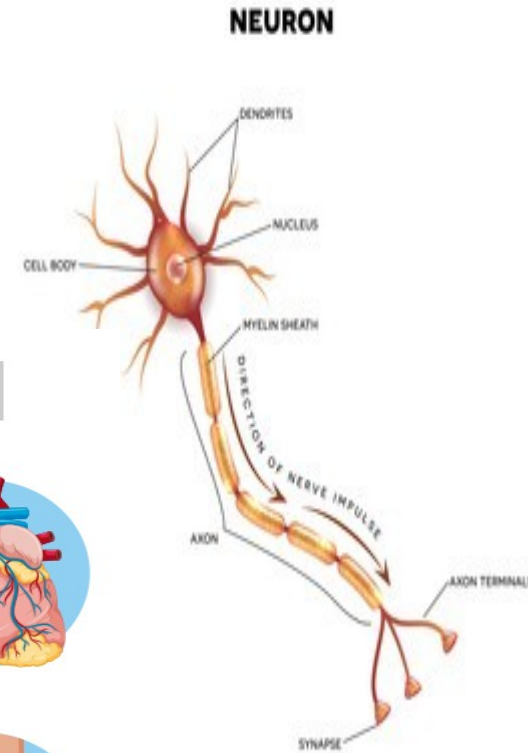
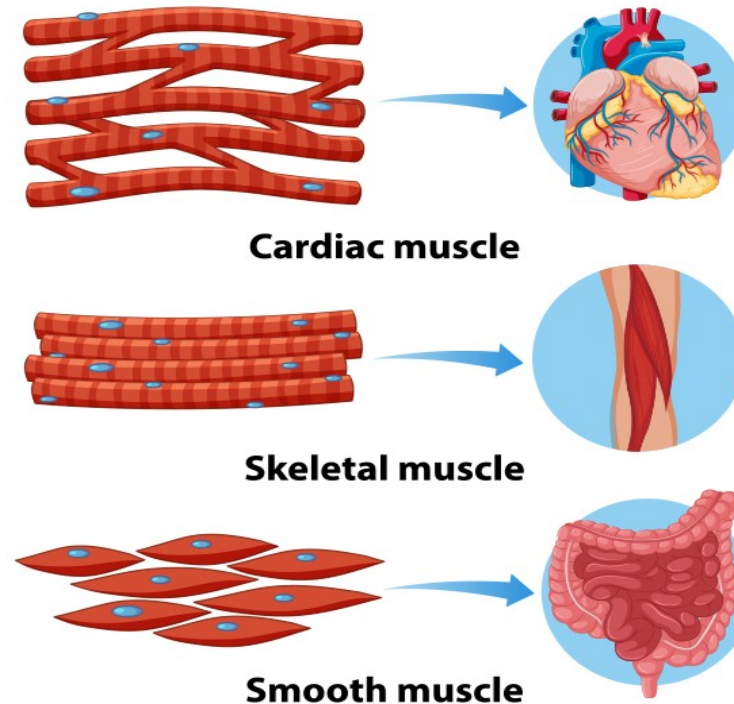
Matrix (ground substance/protein fibers) with special cells. Wide variety.....
Connective tissue



Epithelial – lines hollow organs: heart, stomach/vessels/GI ... covers body: skin

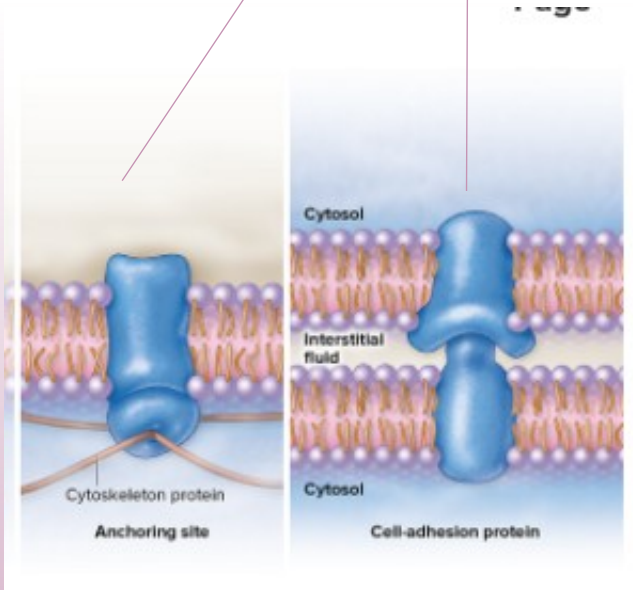
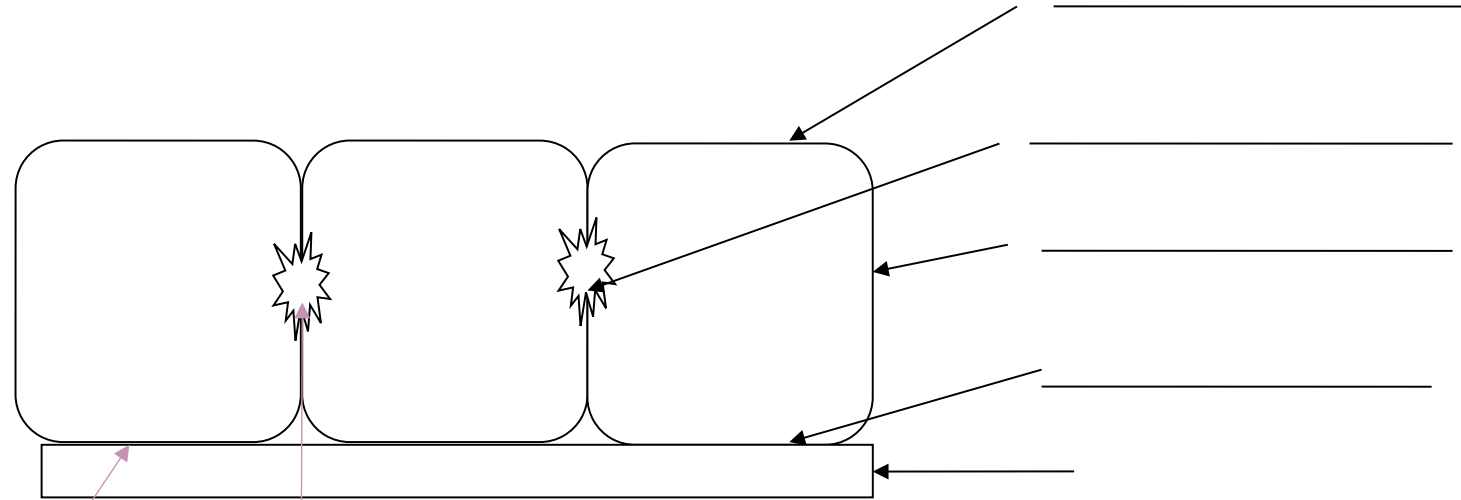


Type of Muscle Cells



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EPITHELIAL



Epithelial Tissues

Characteristics

- Cellular (lots of cells)
- Polar (not a charge – different job on apical versus basal)
 - Attached – stuck to basement
 - Avascular –
- Regenerative – divides at a high rate than any other tissue.

Functions

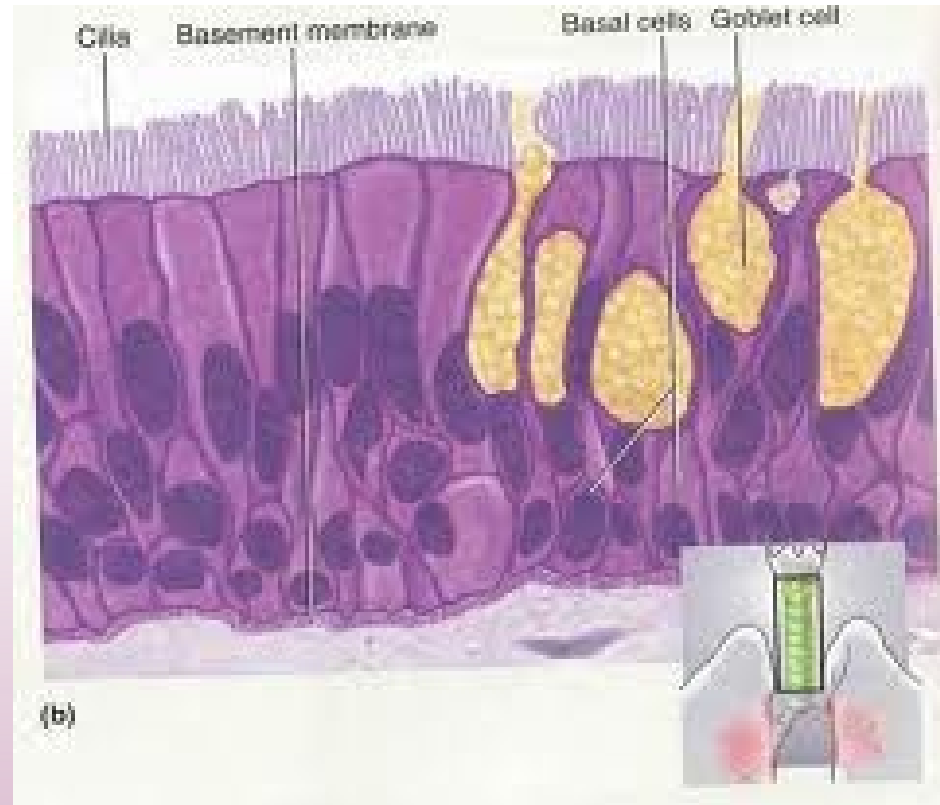
- Protection – stratified
- Permeable (control) – simple squamous – gas/nutrient diffusion
- Sensations - some have lots of nerve ending
- Secretions – all glands – cuboidal!
- Specialized columnar cell – goblet – produces mucus

Epithelial Tissue Type	Location WHY?/Job!
Simple Squamous	
Stratified squamous	
Simple Cuboidal	
Simple Columnar	
Pseudo-stratified Columnar	
Transitional	

Ciliated simple columnar with goblet cells
(yellow)

Pseudo-stratified Columnar Epithelium

AIRWAYS: trachea, bronchial tree (NOT in alveoli)



Questions

Your client states they feel 'short of breath'. Which epithelial tissues might be involved? **Choose all that apply**

- A – ciliated columnar
- B – simple columnar
- C – simple squamous
- D – stratified squamous
- E – simple cuboidal

All Connective Tissues have three things!

- Specialized cells
- Extracellular protein fibers
- Ground substance

Matrix =

Protein fibers AND ground substance

In some tissues there are more fibers than cells – Areolar, reticular, dense reg and irregular, elastic

In some tissues there are more cells – adipose

In some tissues there is more ground substance – Areolar, Blood, lymph

In bone and cartilage there is more matrix – cartilage is tough but flexible – bone is hard with little bit of flexibility. Only a few cells.

Connective T. Proper:

- Loose
 - Areolar, adipose, Reticular
- Dense
 - Reg, irreg., elastic
 - Syrupy ground substance

Fluid Connective:

- Water based matrix

ONLY TWO:

- Blood
- Ly

Supporting Connective Tissue:

- Two – three cell types only
- Matrix is dense/hard
- Cartilage and bone



Tissue	Examples and Where? WHY?
<u>Loose Connective Tissues</u>	
Areolar	
Adipose	
Reticular	
<u>Dense Connective Tissue</u>	
Regular	
Irregular	
Elastic	
<u>Supporting</u>	
Cartilage/Bone	
<u>Fluid</u>	
Blood/lymph	

MEMBRANES

Two main **tissue types** found in membranes:

Epithelial (simple or stratified) and Areolar

Cover/Lines surfaces, or separate organs/lobes

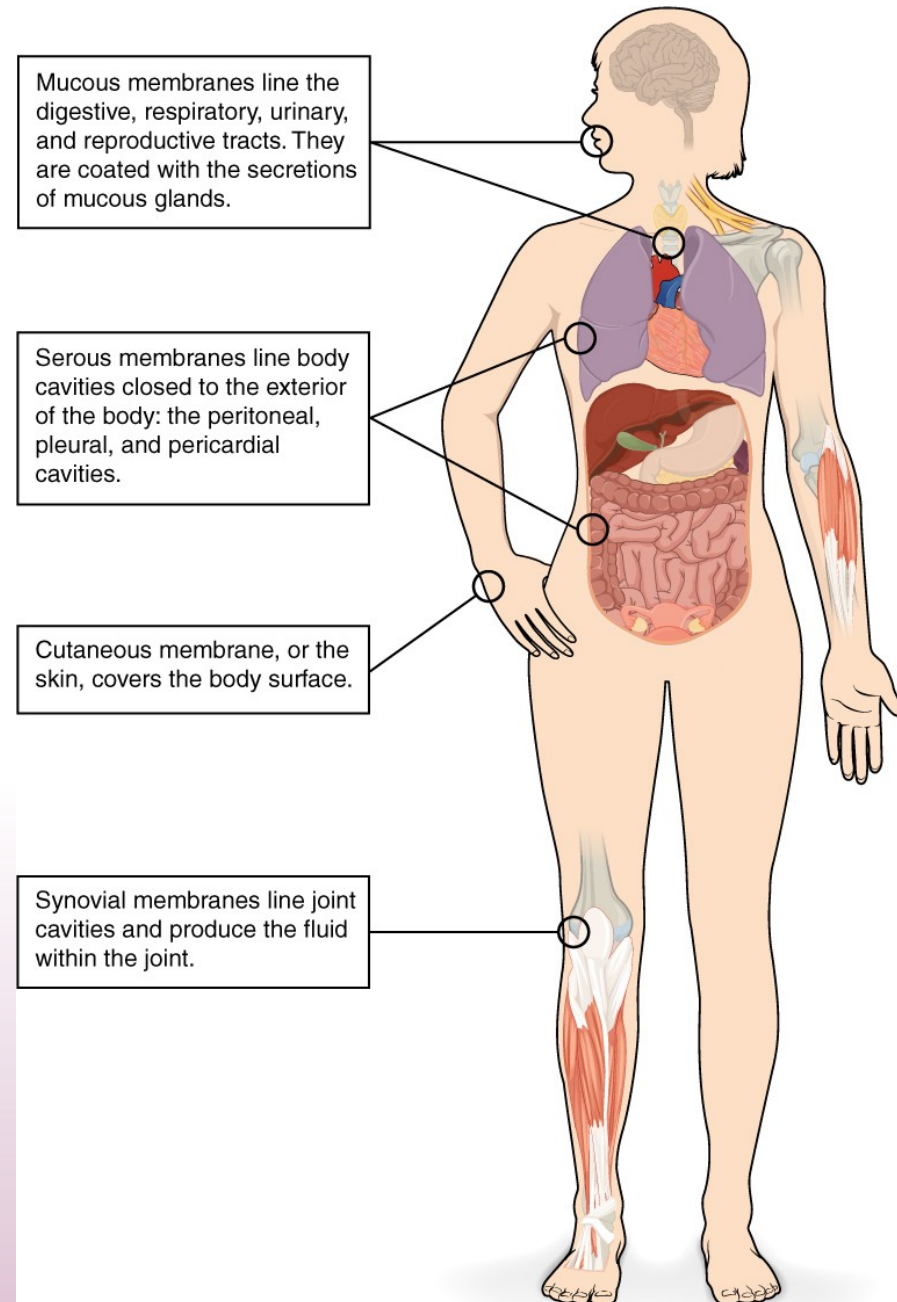
Types of Membranes:

Mucous – lines openings to exterior.

Serous – prevents friction, surrounds heart, lungs, GI – double layer

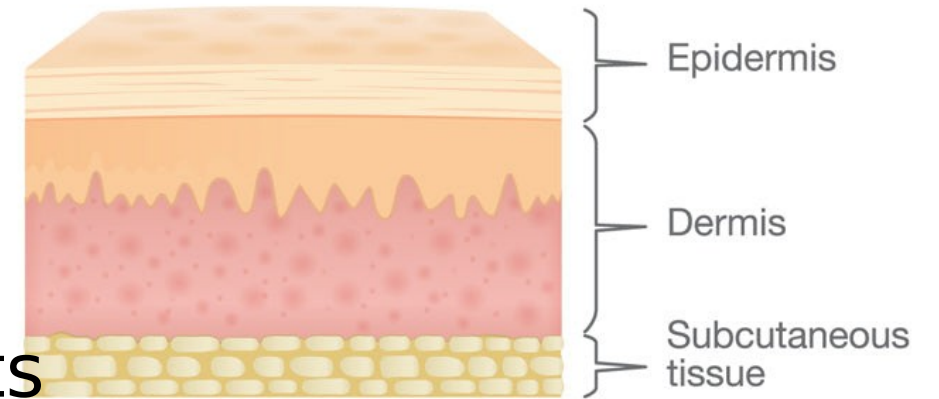
Cutaneous – SKIN - protection

Synovial – lubricates joints



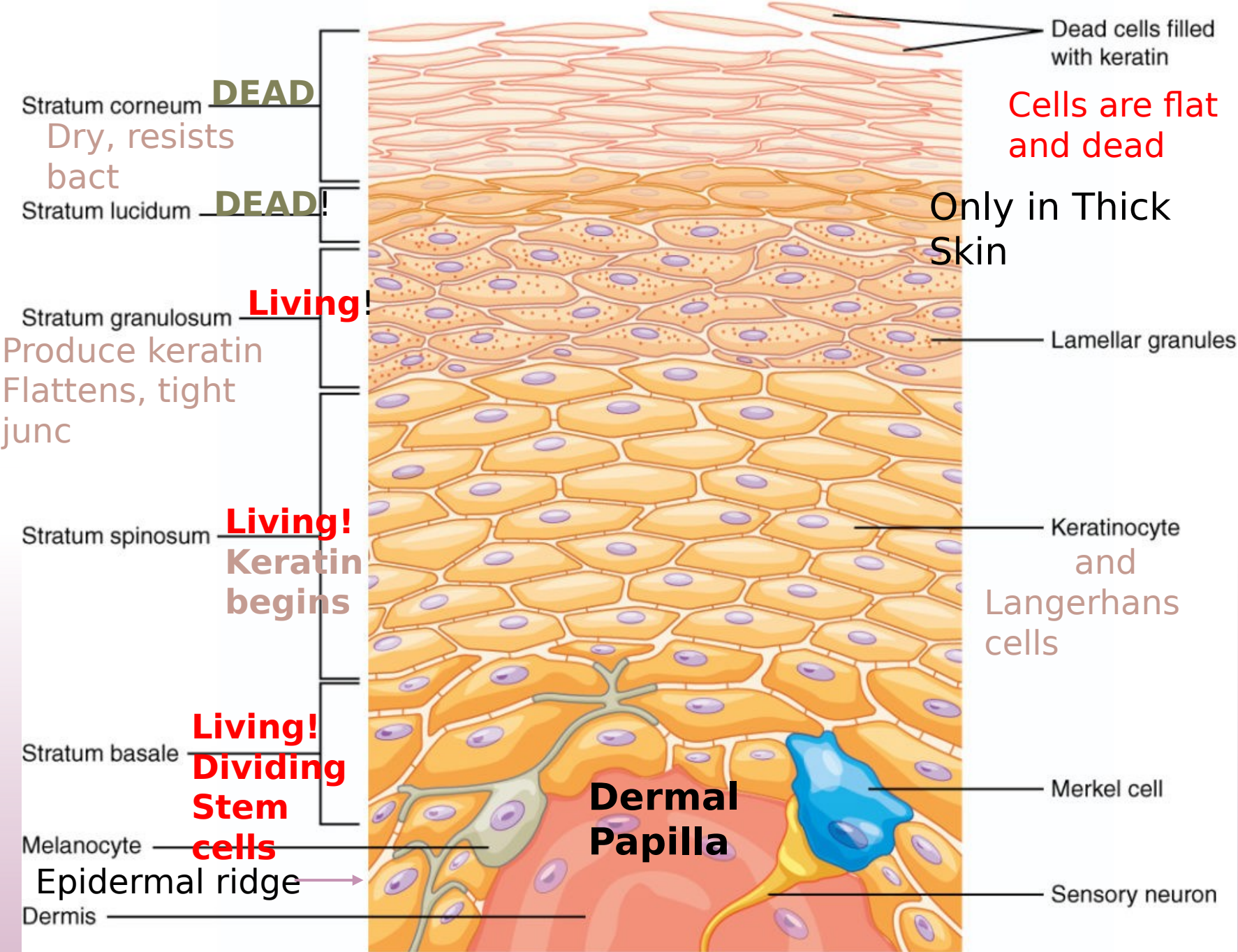
The Layers of Skin

SKIN/INTEGUMENTARY FUNCTIONS



- Protection – environmental threats
 - Melanin – UV light (mutations to DNA)
- Excretion – wastes, salts....
- Helps maintain internal temperature
 - Vasodilation and sweating, vasoconstriction
- Water Resistant – tight junctions, keratin
- Vit D synthesis – details coming up....
- Sensory reception – lots of nerve endings/receptors

Keratinocytes move from Basale layer to corneum. As they move, they flatten and die and fill with keratin.

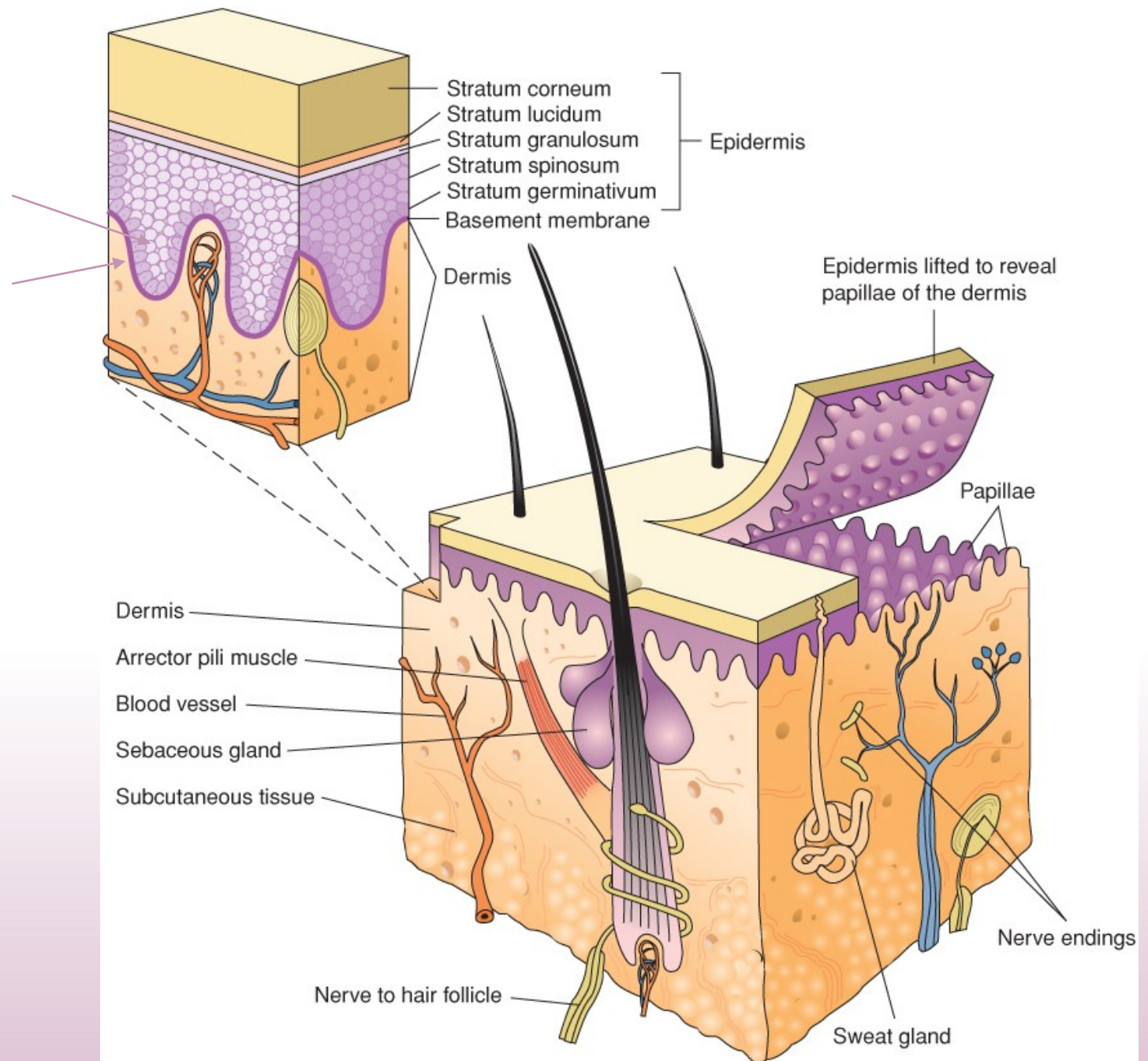


DERMIS

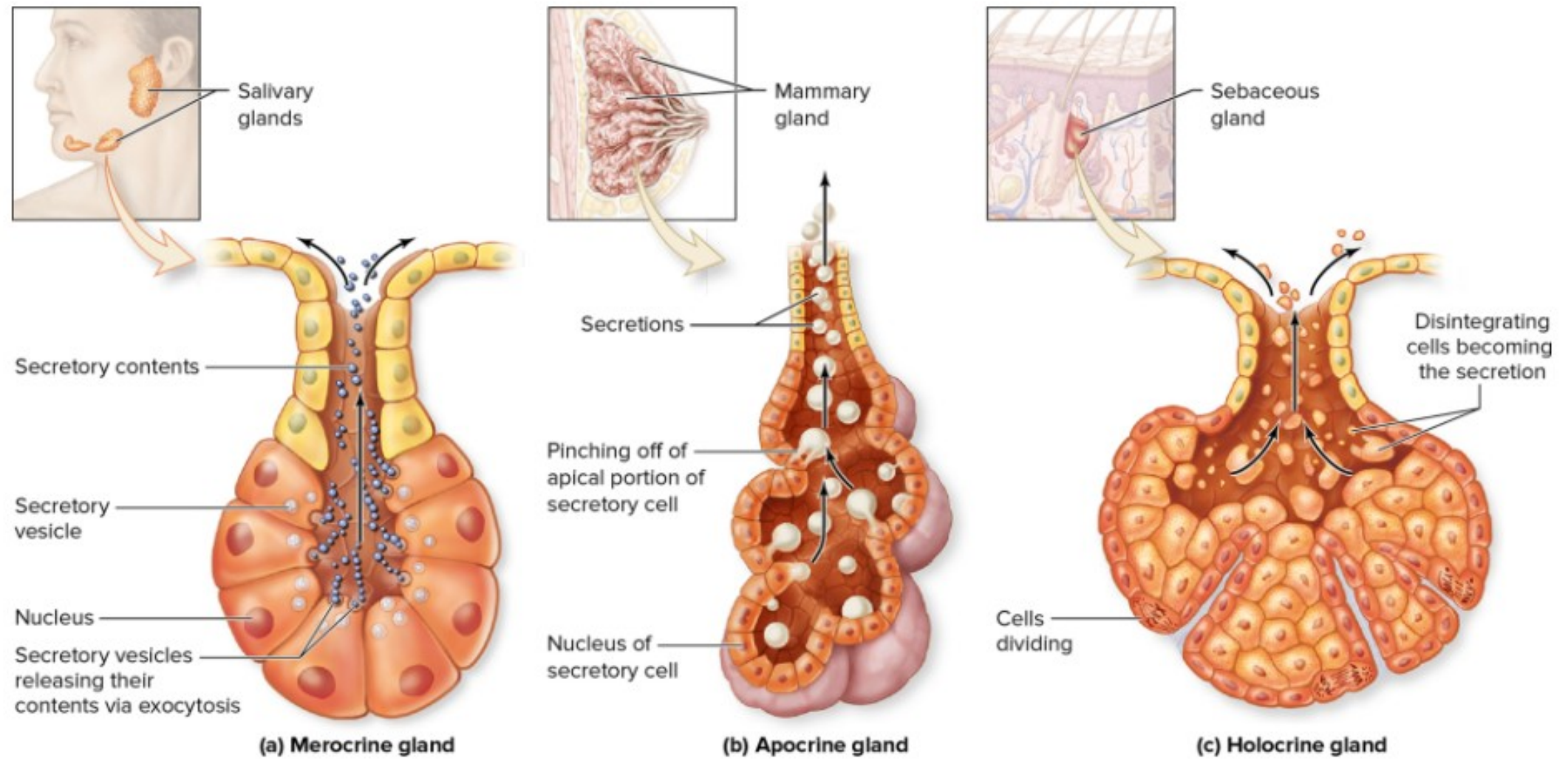
Epidermal
ridges

Dermal
papillae

Note:
No structures in
Epidermis!
LOTS in the
dermis!
Nerves, blood
vessels,
glands.....



Gland types: Exocrine!

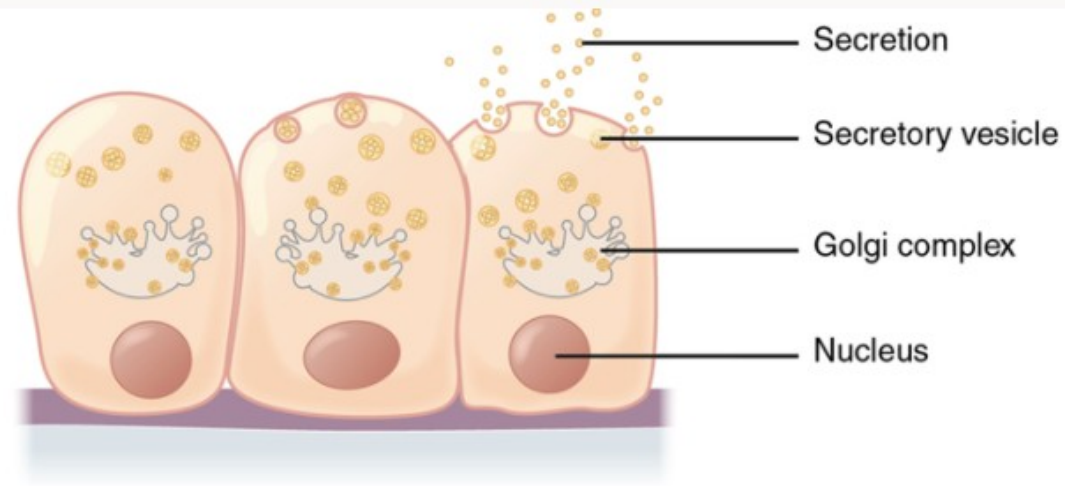


Me=messy - just water

AP=a piece breaks off
Thicker than

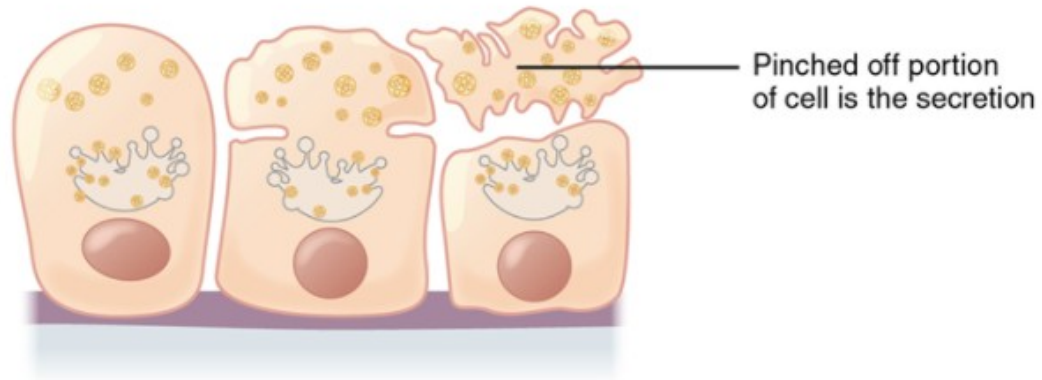
HOL=whole cell bursts
Very thick, oil/fat based!

Merocrine

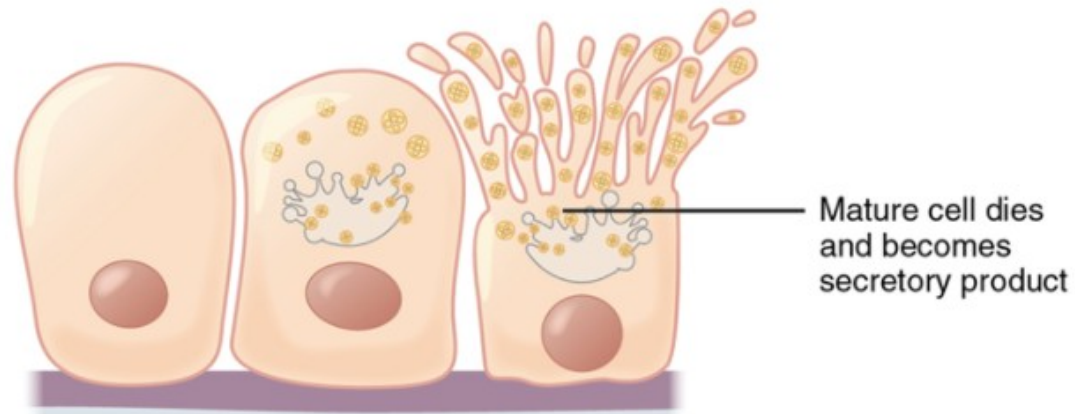


Unicellular –
No need to have
more as the cells
stay intact.

Apocrine- apical
surface breaks off
with secretions



Holocrine

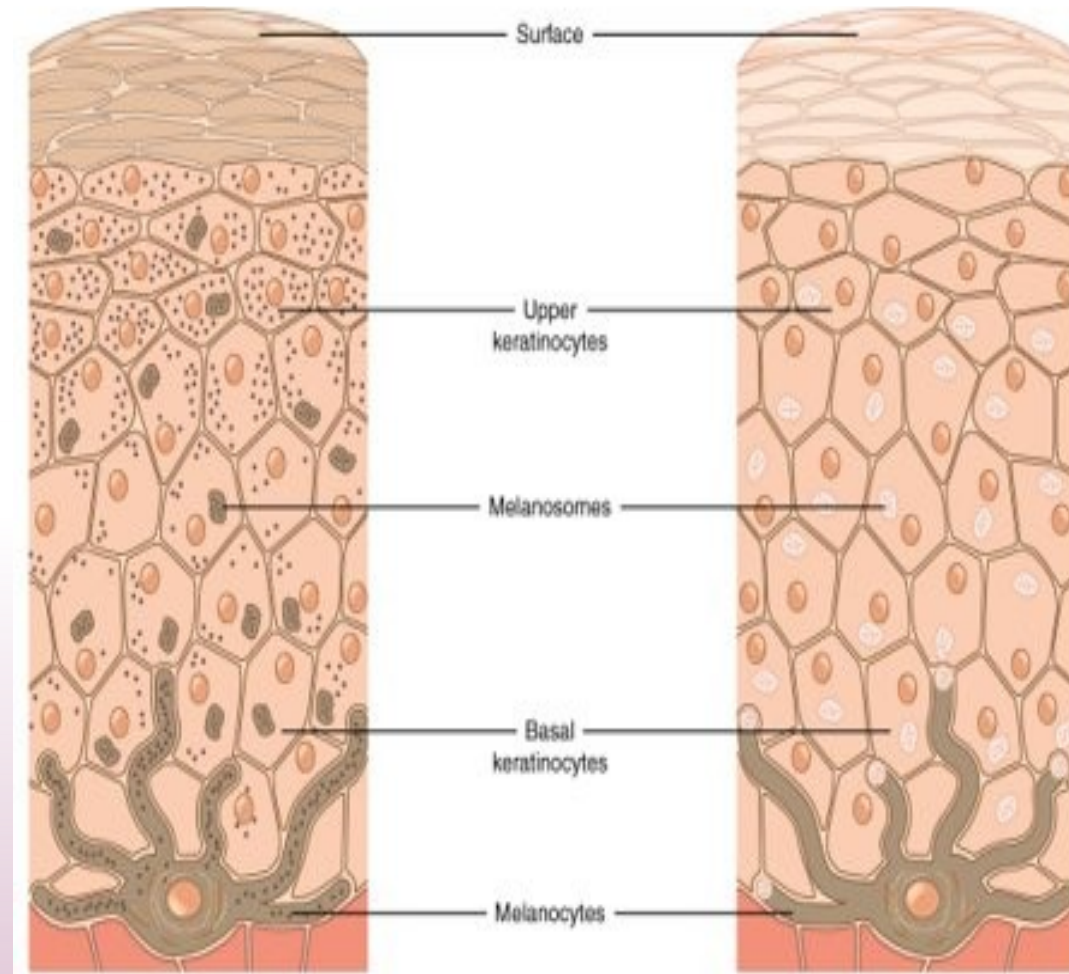


Multicellular –
these cells need
more to replace
them – so
stratified – as a
portion or all of
the cell is
excreted.

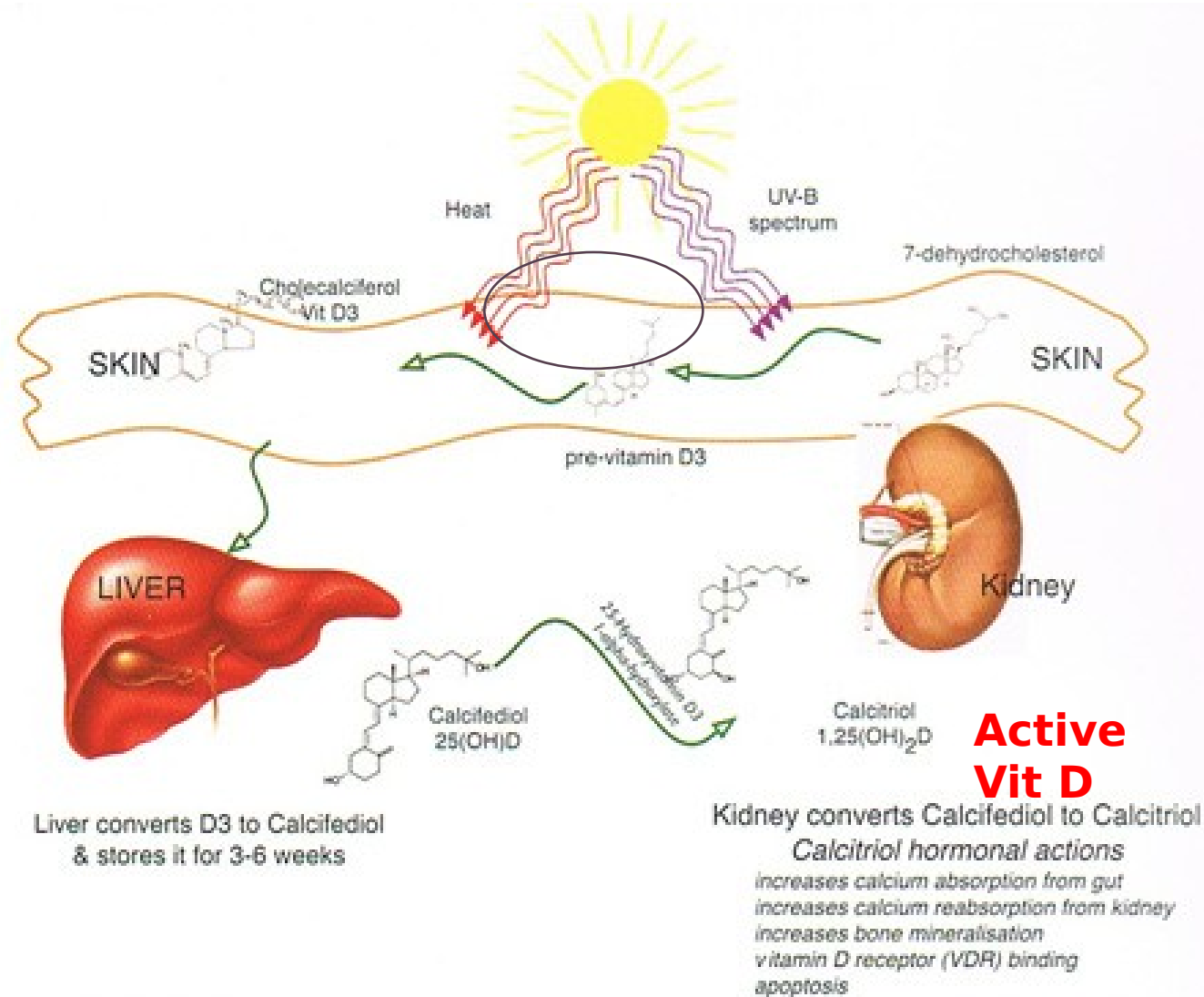
MELANIN - TRIGGERED BY **UVB** RAYS, PROTECTS SKIN DNA CELLS

Melanin is produced
by **Melanocytes**.

- Everyone has the same number.
- They produce DIFFERENT amount and color.



VITAMIN D CONVERSION - NEED UV SUNLIGHT



STEPS:

1- Cholecalciferol from UVB light and 7-dehydrocholesterol (f=first)

2 - liver converts to Calcifediol (di=2)

3- Kidney releases Calcitriol (tri = 3)

OUTCOME:

- 1- GI absorbs more Ca⁺
- 2- Kidneys return Ca⁺ to blood
- 3- Osteoclasts break down bones - release stored Ca⁺

Reminders:

DUE WEDNESDAY: 1st discussion post

DUE SUNDAY:

- Pre-lesson checkpoint
- Module 3 quiz (Respondus)
- Peer response in discussion
- Lab quiz (Respondus)

You can avoid feeling overwhelmed on Sunday by completing something each day - this also leads to better retention of knowledge.